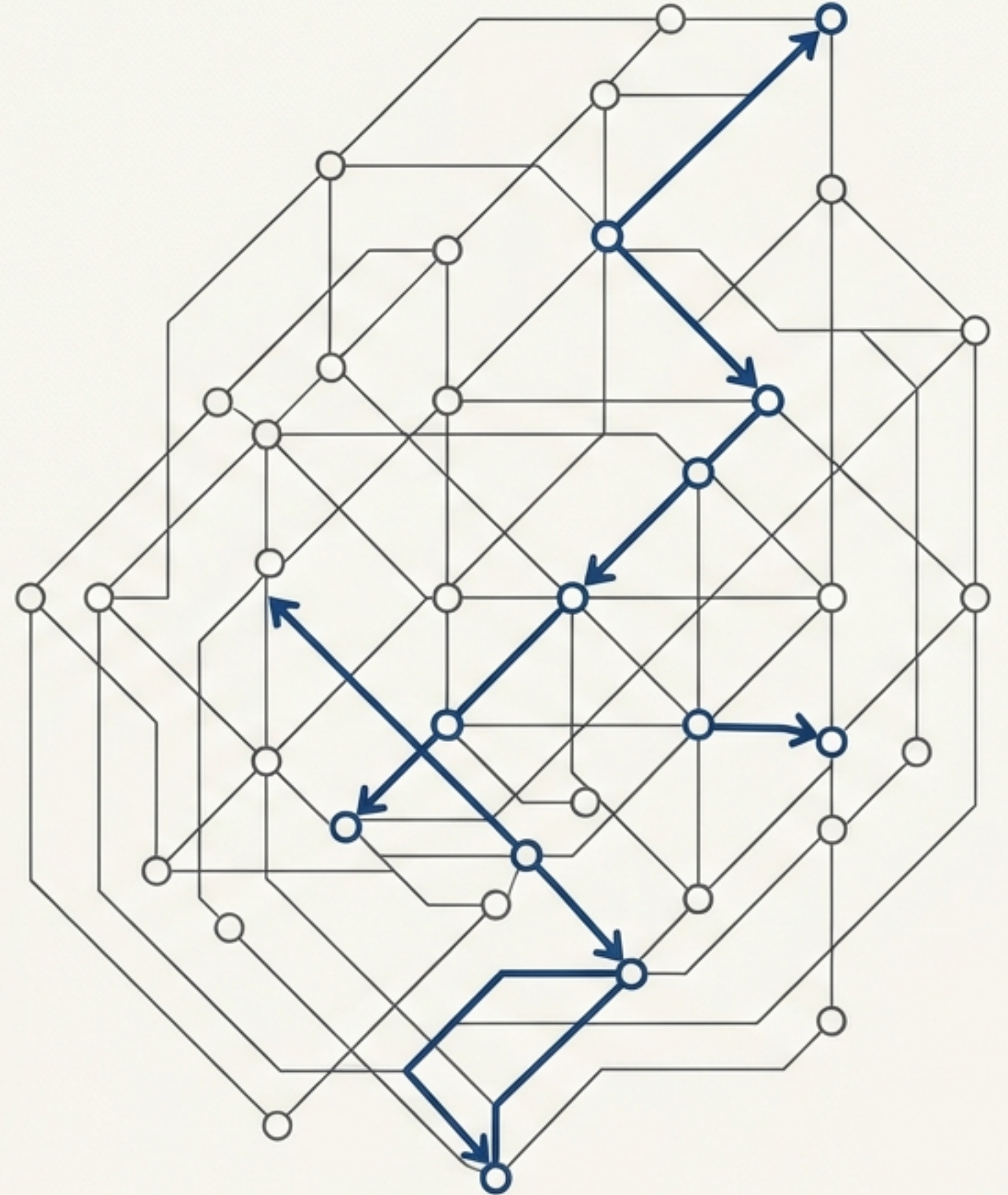


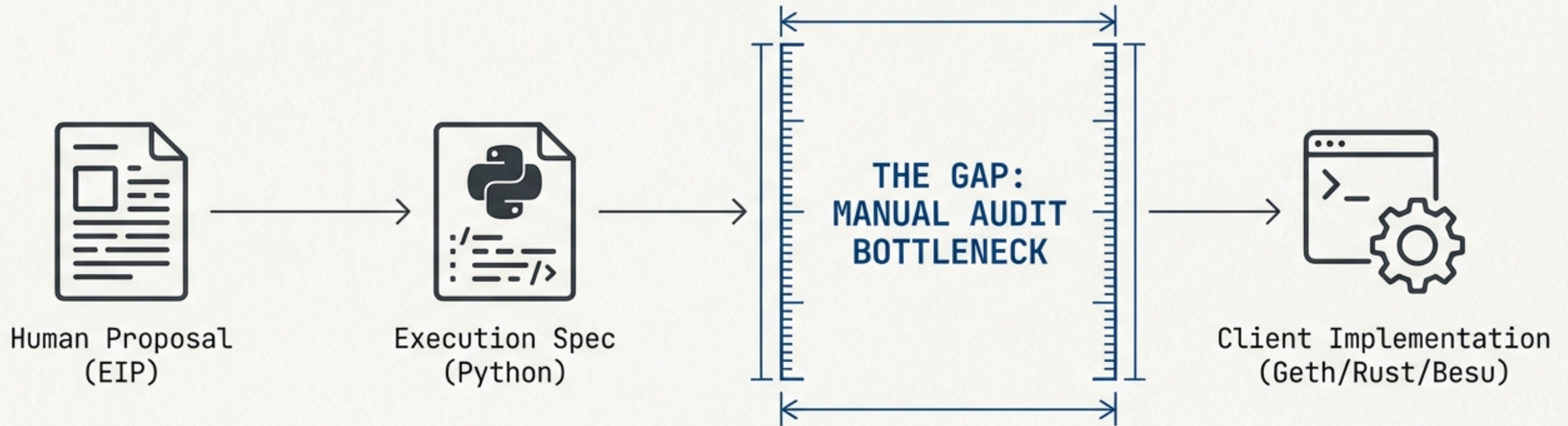
Rooted Layers: AI Verification & Evidence Governance

Insights from the EIP-Verify Project
on Ethereum Protocol Security

FIELD REPORT: LLM-ASSISTED AUDIT SCAFFOLDING // STATUS: COMPLETED

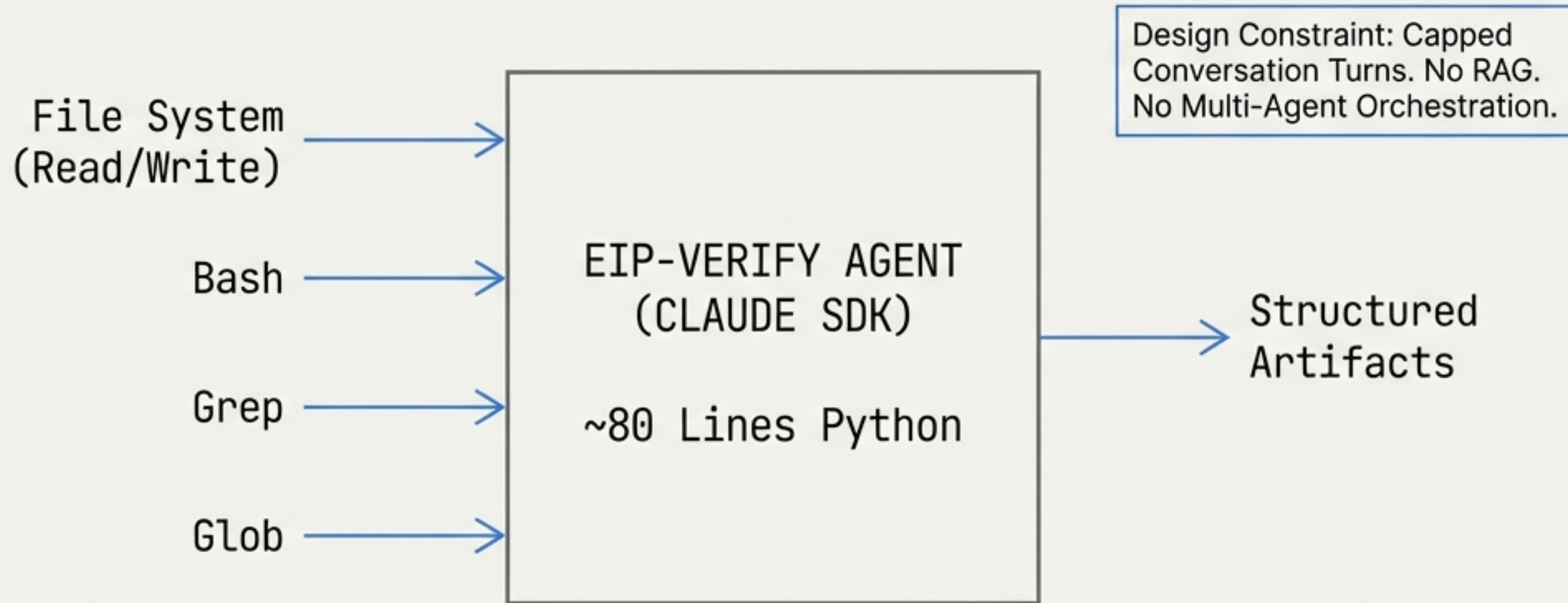


Every Blockchain Runs on a Promise That Code Matches Spec



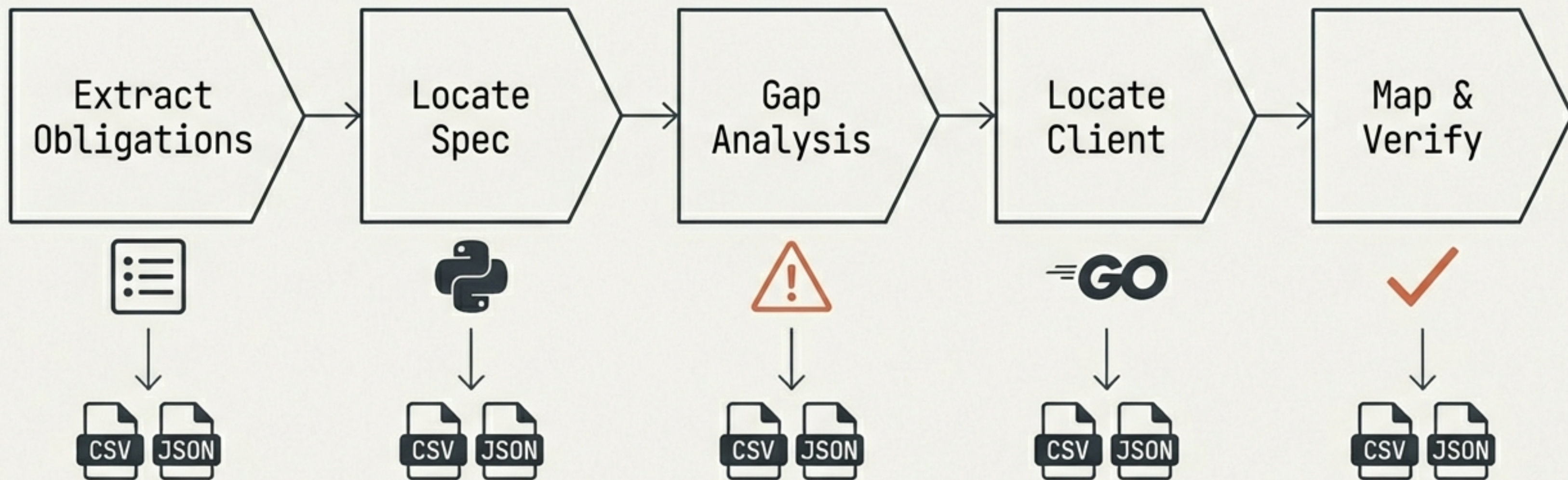
Current State: Requires slow, expert-level manual review to ensure obligations line up across languages and logic.

Complexity Obscures Errors, Simplicity Exposes Them



We avoided black-box frameworks. The criterion was simple: when the output is wrong, can you figure out where and why?

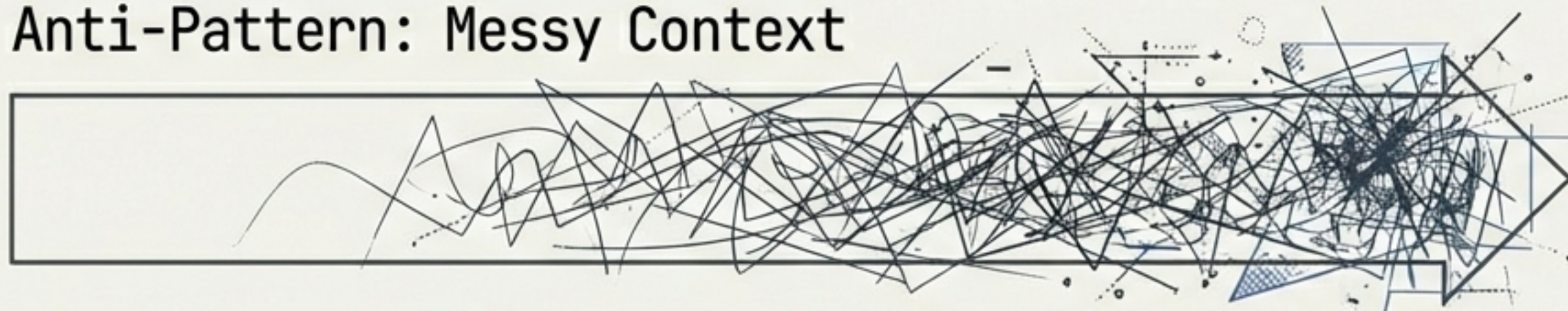
The Five-Phase Verification Pipeline



System stress-tested on EIP-1559 (Fee Market) and EIP-7702 (Prague-fork). Each phase produces a discrete, inspectable artifact.

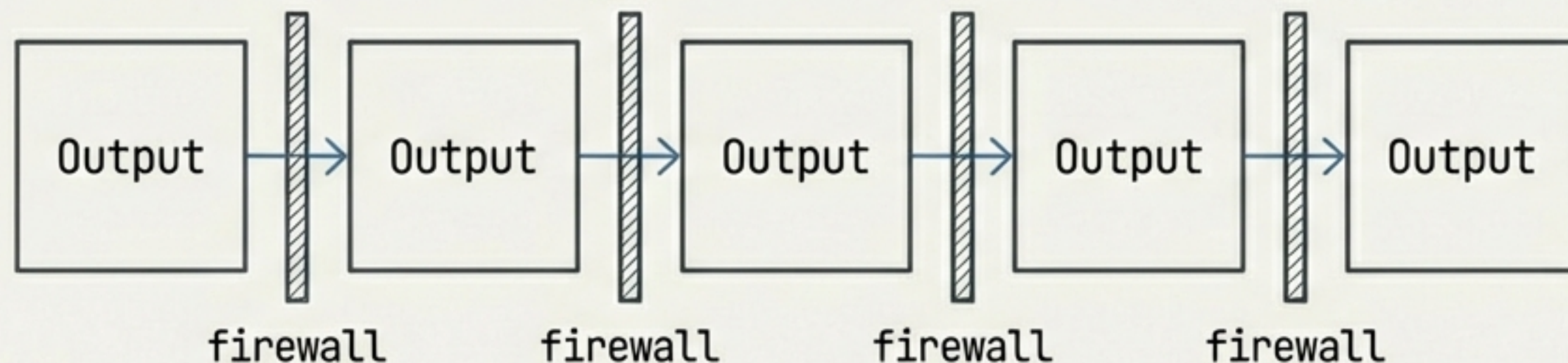
Why Phase Boundaries Are Critical for Trust

Anti-Pattern: Messy Context



Errors compound. Hallucinations in Step 1 bias Step 5.

Rooted Layers Design: Clean Context



Fresh context per step. Errors are quarantined.

- Benefits
 1. Fresh Context (No compounding)
 2. Constrained I/O Contracts
 3. Single-phase Replayability

The Failure Mode is Plausible-Sounding Wrongness

Case Study: CTR-001 (EIP-1559)

Analysis Complete. All obligations properly mapped to specification and client implementation code.

Status: Verified.

ID: OBL-05 | Spec: valid | Client Location:
'accounts/abi/bind/bind.go', 'ethclient/test.go'

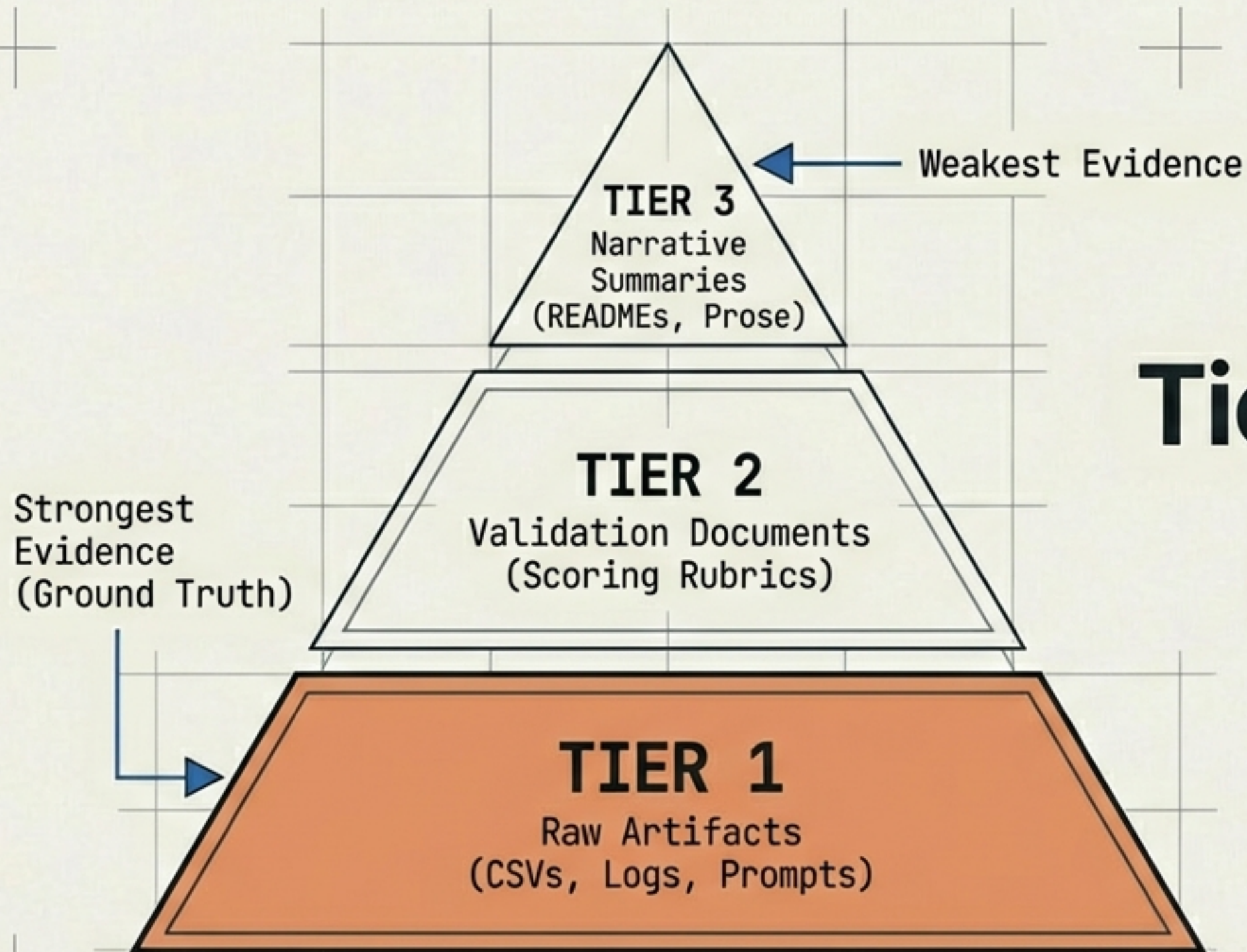
IRRELEVANT TEST FILE

The system didn't output gibberish.
It output a perfect summary that contradicted its own data.

The system got dramatically better at exposing where it might be wrong. It did not get dramatically better at being right.

The gap between auditability and correctness became the most important finding of the project.

Governance Tier 1: The Evidence Hierarchy



Tier 1 > Tier 2 > Tier 3

This rule prevents confident narratives from overriding contradictory data.

Governance Tier 2: The Scoring Rubric

Dimension	Score (0-2)	Criteria
Statement Fidelity	[] [] [x]	Matches EIP text?
Spec Location	[] [x] []	Relevant Python code?
Client Location	[x] [] []	Relevant Implementation?
Flow Plausibility	[] [x] []	Connects entry to logic?
Evidence Sufficiency	[] [] [x]	Data allows judgment?

Scores map to labels: Valid (≥ 8), Partial (5-7), Invalid (≤ 4), Disputed.

Governance Tier 3: The Contradiction Register

| CTR-001 | STATUS: **OPEN** ☒
| CONFLICT: Summary claims verified;
| Data points to test files.

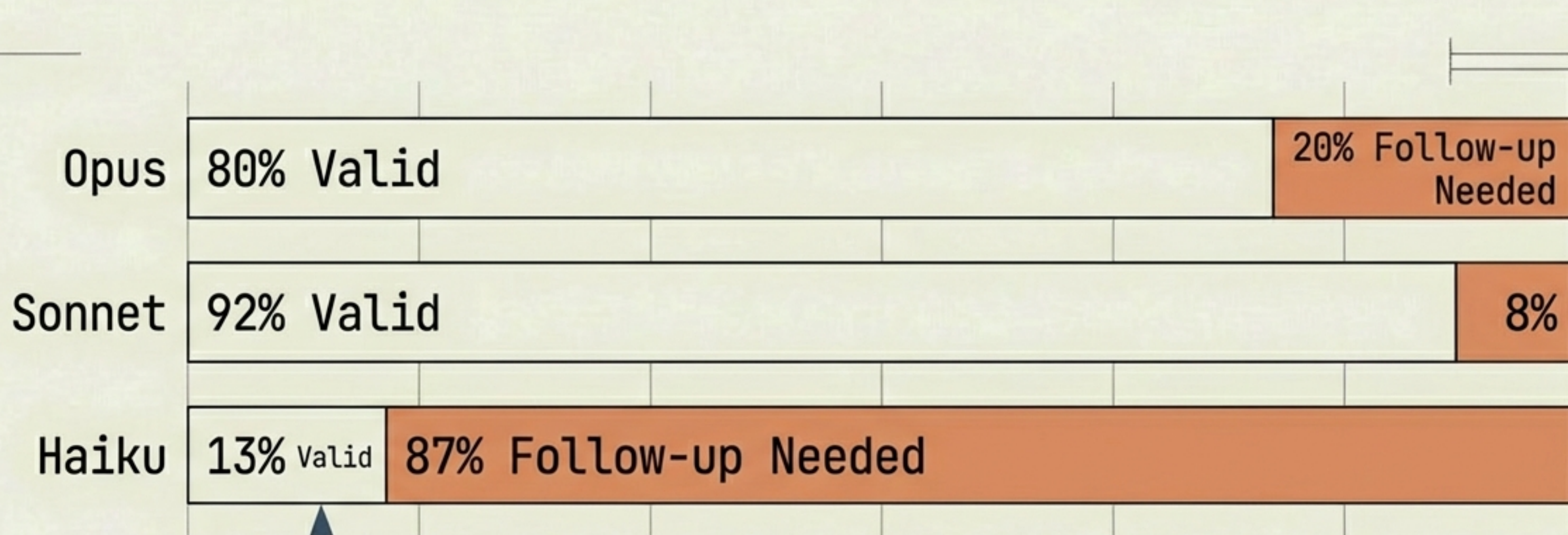
| CTR-002 | STATUS: **RESOLVED** ☒
| CONFLICT: Spec ID mismatch between
| runs.

| CTR-003 | STATUS: **OPEN** ☒
| CONFLICT: Narrative hallucinated
| function name.

**We don't hide bugs;
we catalogue them.**

Contradictions are
treated as first-class
workflow objects to
ensure inspectability.

Model Choice is a Governance Decision

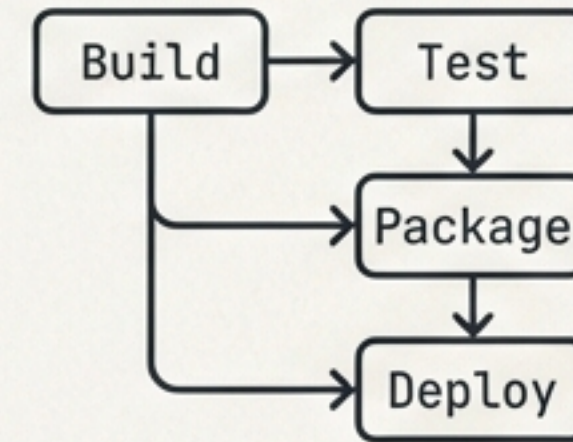


Triage-grade only. 72% of total human verification burden came from Haiku outputs.

Deployment Surface: CI Pipeline Over Chatbots



Chat Interface
(Variable Inputs, Ephemeral)



CI Pipeline
(Fixed Inputs, Immutable Artifacts)

- Fixed EIP # and Client Version
- Reduced Attack Surface
- Immutable Output Bundle

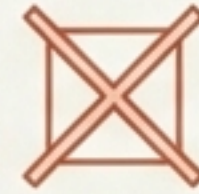
For security tooling, flexibility is a liability. We need reproducible artifacts.

The Verdict: Force Multiplier, Not Replacement



Defensible Claims

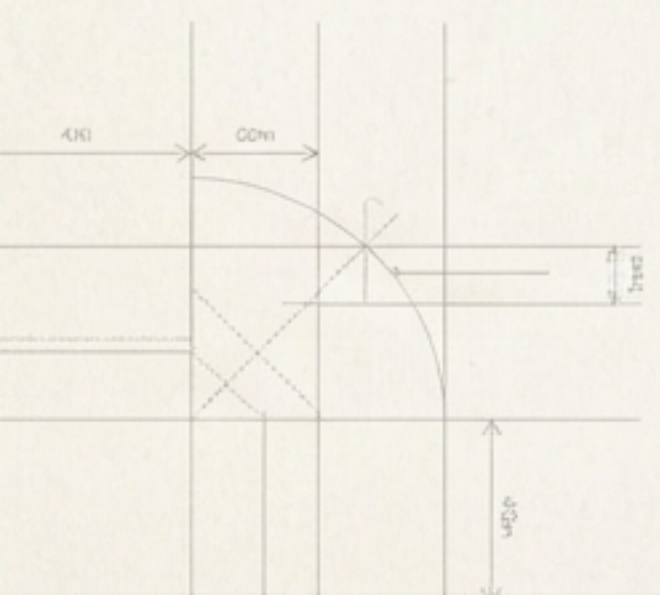
- Accelerates triage process.
- ~80% of mapping pre-done.
- Clear flags for human review.



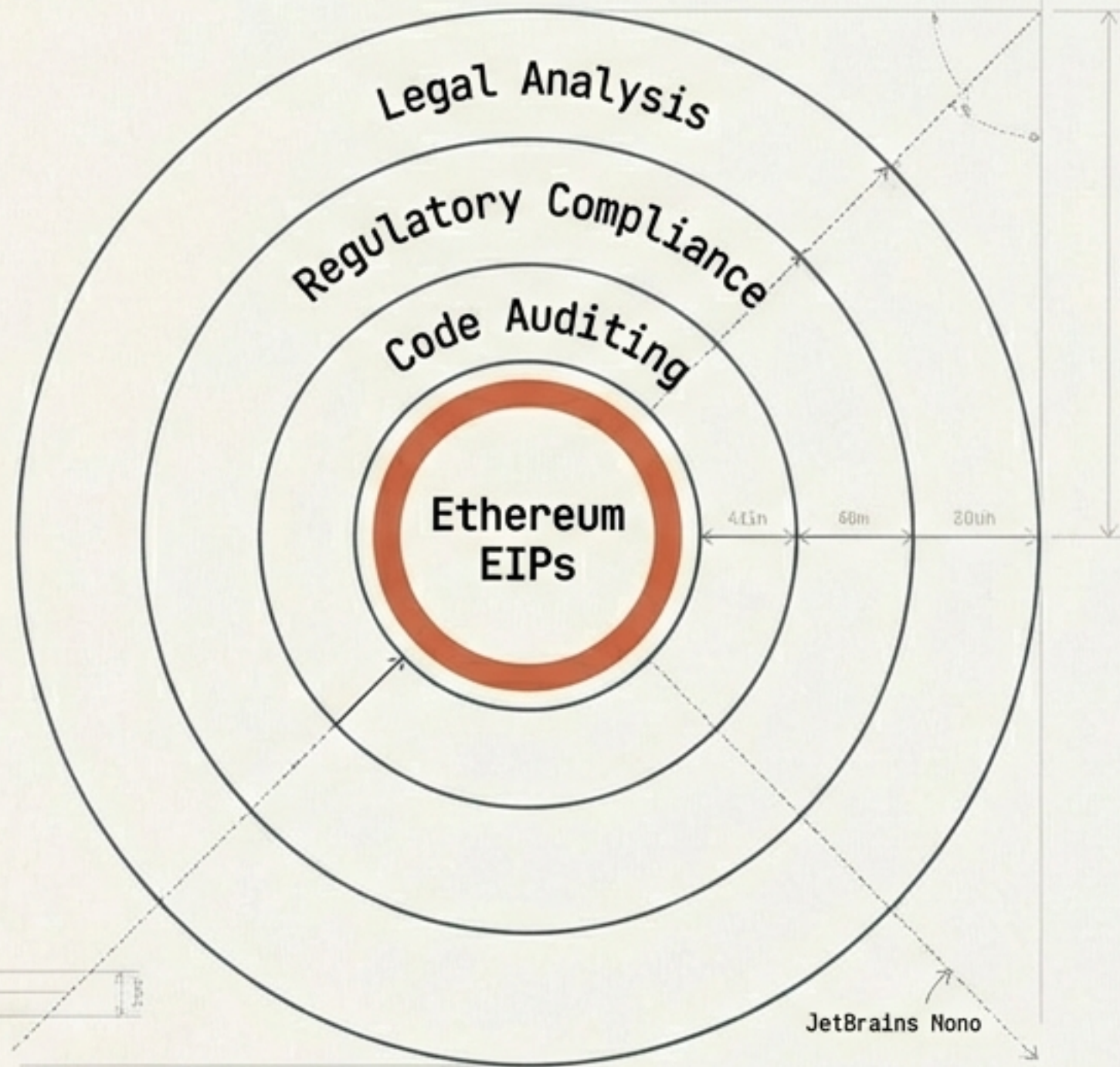
Unsupported Claims

- Autonomous verification.
- "High confidence" without human-in-the-loop.

Opus at 80% valid is operationally useful today for triage acceleration, but **requires the human expert.**



Generalizing the Pattern Beyond Ethereum



Universal Lesson:
Auditability improves
faster than correctness.

The real metric is cost-to-trust,
not cost-to-generate.

The Maturity Signal That Matters

github.com/rooted-layers/eip-verify



The strongest contribution isn't a claim that verification is solved.
It's a workflow that got better at exposing where it might be wrong.